

System-Technology Codesign of 3-D NAND Flash-Based Compute-in-Memory Inference Engine

WONBO SHIM^{ID} and SHIMENG YU^{ID} (Senior Member, IEEE)

School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA 30332 USA

CORRESPONDING AUTHOR: W. SHIM (wshim30@gatech.edu)

ABSTRACT Due to its ultrahigh density and commercially matured fabrication technology, 3-D NAND flash memory has been proposed as an attractive candidate of inference engine for deep neural network (DNN) workloads. However, the peripheral circuits require to be modified with conventional 3-D NAND flash to enable compute-in-memory (CIM), and the chip architectures need to be redesigned for an optimized dataflow. In this work, we present a design of 3-D NAND-CIM accelerator based on the macro parameters from an industry-grade prototype chip. The DNN inference performance is evaluated using the DNN+NeuroSim framework. To exploit the ultrahigh density of 3-D NAND flash, both inputs and weights mapping strategies are introduced to improve the throughput. The benchmarking on the VGG network was performed across the technological candidates for CIM, including SRAM, resistive random access memory (RRAM), and 3-D NAND. Compared to the similar designs with SRAM or RRAM, the result shows that the 3-D NAND-based CIM design can achieve not only 17%–24% chip size but also 1.9–2.7 times more competitive energy efficiency for 8-bit precision inference. Inference accuracy drop induced by 3-D NAND string current drift and variation is also investigated. No accuracy degradation by current variation was observed with the proposed input mapping scheme, while accuracy drops sensitive to the current drift, which implies that some compensation schemes are needed to maintain the inference accuracy.

INDEX TERMS 3-D NAND, compute-in-memory (CIM), deep neural network (DNN), hardware accelerator.

I. INTRODUCTION

SINCE the state-of-the-art deep learning algorithms tend to grow the larger size and deeper network to achieve high accuracy, the tremendous number of computations is required. Therefore, a huge volume of data movements between the processor and off-chip memory in conventional von-Neumann architecture (e.g., GPU) significantly lowers the energy efficiency for deep neural network (DNN) workloads. There have been lots of efforts from industry and academia to propose alternative computing platforms. Several CMOS-based domain-specific accelerators, such as Google TPU [1], were designed to alleviate the problem by data reuse on-chip, but the memory wall problem still remains where the model parameters are stored in the global buffer, and the actual computation is performed at the digital multiply-and-accumulation (MAC) arrays.

To overcome these challenges, compute-in-memory (CIM) has been emerged as an alternative paradigm due to its high throughput and energy efficiency [2]. CIM utilizes the conductance of the memory cell to represent the weight and conduct MAC operations by activating multiple rows and reading out the analog current summed up along the column. Emerging nonvolatile memory (eNVM) devices, such

as resistive random access memory (RRAM) [3], [4] and phase-change memory (PCM) [5], have been investigated as synaptic devices for vector-matrix multiplication (VMM) or weighted sum computation due to its logic compatibility and nonvolatility. However, their relatively small ON/OFF current ratio and large ON-current are not suitable for large array configuration. Floating gate type NOR flash [6] technology is another candidate due to its large ON/OFF current ratio, but the large ON-current still makes sense amplifier design of summed readout current exceedingly challenging. Moreover, embedded NOR flash structure is hard to scale down beyond the 28-nm node, so it is less competitive than other device technologies in terms of memory density.

Very recently, NAND flash has been proposed as a high-density and high-bandwidth CIM candidate [7], [8]. Since 3-D NAND flash has the highest density among all the memory devices [9], the weights of large-scale DNN can be stored in a small form factor. Furthermore, it is quite advantageous that 3-D NAND flash is already matured today and is based on a widely commercialized fabrication technology.

This article is an extension of our conference paper [10]. In this work, we present the architectural design of a 3-D NAND flash-based CIM accelerator that is optimized to the

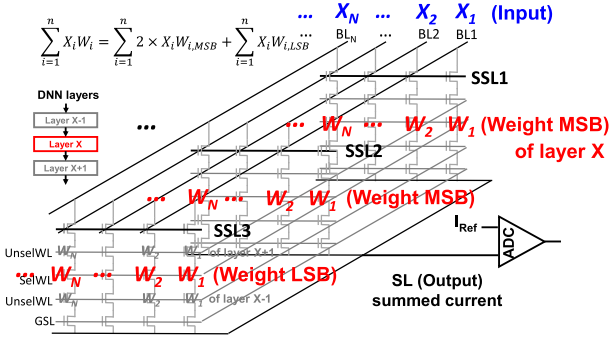


FIGURE 1. VMM operation in 3-D NAND flash in one block with MSB weight decimal mapping.

inference of DNN, with the benchmarking results using the DNN+NeuroSim [11] framework. We used the electrical parameters and the physical dimensions of a 3-D NAND-CIM prototype by Macronix [8] for the baseline of this work. This prototype of 3-D NAND is of industry-grade and has been customized to support the CIM paradigm. The peripheral circuits and chip-level hierarchy configuration have been adapted to support DNN models, such as VGG [12] for image classification. We report the performance (energy efficiency and throughput) results across technological options for CIM. Finally, NAND string current variation caused by threshold voltage variation and back pattern dependency is incorporated into the software simulation to investigate the inference accuracy degradation on the proposed architecture.

II. DESIGN OF 3-D NAND-BASED INFERENCE ENGINE

A. 3-D NAND ARRAY-LEVEL DESIGN

Fig. 1 shows the VMM operation with the schematic of the CIM in 3-D NAND flash memory of which the array structure is exactly the same as the commercial 3-D NAND flash. Prior flash-based CIM topologies [6], [7] activate the WLs as the input vectors of MAC, while the bitlines (BLs) are activated as input vectors in this work similarly as proposed in [8] on single-gate vertical-channel (SGVC) structure. Although both WL input scheme and BL input scheme work properly in terms of functionality, the WL input scheme in 3-D NAND would suffer from huge energy consumption due to the large WL capacitance of the 3-D NAND array and frequent charging and discharging of the WLs as input vectors. The BL input scheme can minimize the WL charging energy, especially in convolutional neural networks (CNNs), because WL charging is required only once for a single layer of CNN.

For the BL input scheme, the multiple weights ($W_1, W_2, \dots, W_N$ in Fig. 1) are stored in different BLs. Data 0 is programmed to have a high threshold voltage, and data 1 is programmed to have a low threshold voltage. Both states are programmed with write-verify to fine-tune the threshold voltage. During the MAC operation, while the BL voltages are applied to the multiple BLs respective to the input vectors, the read voltage is applied to the selected WL of the NAND string, and the pass voltage is applied to the unselected WLs (i.e., the read-out is performed WL by WL). Eventually, the currents flow only through the strings with BL input data 1 and weight data 1 (low threshold voltage) in selected WL. Because the SL of the NAND strings within a single block

is entirely connected through the bottom substrate, the total summed current can be sensed at the end of the SL of the block. Since NAND flash has an exceptionally large page size (several kB of BLs) in a single block, this topology has an advantage in that the huge number of string currents can be summed at once using a single analog-to-digital converter (ADC).

We designed the 3-D NAND block having three SSLs to store the 2-bit weight and being computed at once. The most significant bit (MSB) of weights is stored in the cells of SSL1 and SSL2, while the least significant bit (LSB) of weights is stored in SSL3 only. Since the SL is connected to all SSL together, the output current follows the equation in Fig. 1, which means that 2-bit weights are computed in an analog manner without the digital adder and shifter. By using this MSB weight decimal mapping, the total computing operation steps decrease by half.

The electrical characteristics of NAND flash device of our work are based on the measured experimental data of a 32-layer 3-D NAND chip reported in [8]. The 3-nm-thick polysilicon channel could have extremely low ON-current (~ 2 nA) and OFF-current (below 1 pA) due to the low mobility and high ON/OFF ratio of NAND flash device. The low ON- and OFF-currents are suitable for activating the very large number of BLs at the same time with proper summed current to design the ADC.

To determine the physical dimensions of the 3-D NAND array for one block, we used the standard 3-D NAND pitch size [13], i.e., BL pitch: 40 nm and SSL pitch: 0.75 μm . The number of BLs of the array is determined according to the kernel size of the largest convolution layer of the DNN to unroll each kernel and map it into a single array. The kernel size of the largest layer of VGG-8 is $3 \times 3 \times 512$; then, we will map the weights to other BLs three times as will be explained in Section II-B, so one block has a total of 13 824 BLs and 553- μm width. Along the vertical direction, one block consists of three SSLs, as shown in Fig. 1, so the height of a block is 2.25 μm . As a result, the number of the physical cells in one block is 13 824 BL $\times$ 32 WL $\times$ 3 SSL, i.e., 1.27 Mb. This block size configuration follows the general design guidelines of the commercial 3-D NAND.

B. MULTIBIT INPUT DECIMAL MAPPING

Different from the CIM arrays with other NVM devices, the NAND CIM array has a huge number of inputs (BLs) in a single array, so, in most of the cases, it is larger than the input vector size. Since the remaining empty BLs compose the same block and share the ADC with used BLs, storing (or coping) the weights of different kernels to the empty cells is not meaningful in terms of latency improvement. However, we can copy and map the input and the respective weight of the same kernel to the empty BLs and take advantage of input cycle reduction. Fig. 2 shows the mapping scheme of a convolutional kernel with the MSB weight decimal mapping and n -bit input decimal mapping. The n -bit input can be represented using $2^n - 1$ BLs. For example, 2-bit input can be represented with three BLs using one BL as LSB input and two BLs as MSB input. The weights are mapped to three times more BLs that are sharing the same SSL; the summed-up current at SL is the multiplied result of the 2-bit input and the 2-bit weight. Therefore, the input vector cycles

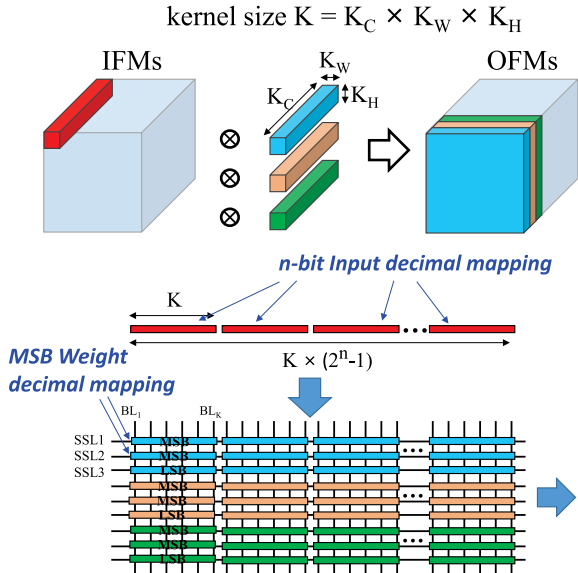


FIGURE 2. Mapping scheme of a convolutional kernel with MSB weight decimal mapping and multibit input decimal mapping.

are reduced by using the multibit input decimal mapping, which means that the throughput can be improved. Depending on the kernel size of the layer, we could map the inputs up to 255 times more BLs for the 8-bit input decimal mapping.

We defined the max-bit input decimal mapping as the n -bit input of a particular convolutional kernel (size $K = K_C \times K_W \times K_H$) is mapped to $K \times (2^n - 1)$ BLs with the largest number n , where $K \times (2^n - 1)$ does not exceed the number of BLs in a designed subarray. In this work, the NAND array is designed to have 13 824 BLs to map 2-bit input of the largest convolutional kernel of VGG-8, i.e., 2-bit input decimal mapping. The max-input decimal mapping could adaptively adjust the copies of input vectors depending on the kernel size and the number of BLs. In the case of fully connected (FC) layers, we used 2-bit input decimal mapping for all the FC layers for simplicity.

C. SUBARRAY ARCHITECTURE DESIGN

The subarray is the minimum unit of the chip hierarchy with multiple 3-D NAND blocks. In our design, the subarray has 64 blocks with 144- μm BL length, which is relatively much shorter than conventional NAND flash (generally several millimeters long) to achieve small RC loading with faster BL charging speed and less energy consumption. Following the method that 2-bit weight is represented in one block, then 8-bit weight needs four blocks to be stored (ADC outputs from the blocks with more significant bits are shifted), so 16 kernels can be stored in a subarray with 8-bit weight. The total capacity of a subarray is 81 Mb.

We estimated the latency and energy parameters of the NAND array with *RC* delay model. Due to the small dimension of WL and BL than the conventional 3-D NAND, the estimated latencies of WL and BL setups are 303 and 12 ns, respectively, as shown in Table 1. We ran the HSPICE simulation to achieve the SL charging time, and the result shows that the summed current for the ADC sensing (in the case of max-bit

TABLE 1. Latency and energy estimation for parallel read in 64 block subarray.

	Latency	Energy
<i>WL setup</i>	303 ns	43.5 nJ
<i>BL setup @ 50% sparsity</i>	12 ns	35.9 pJ (5.2 fJ per BL)
<i>SL setup @ max-bit input</i>	530-750 ns	41.7 pJ

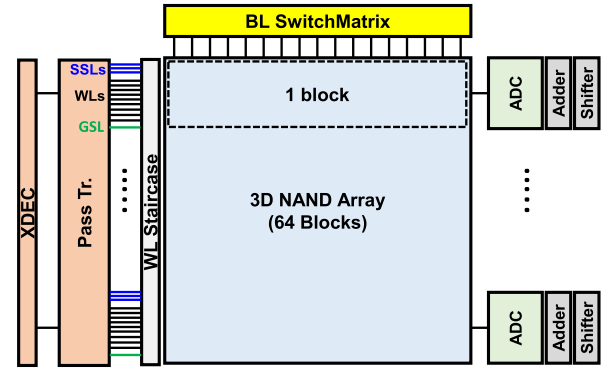


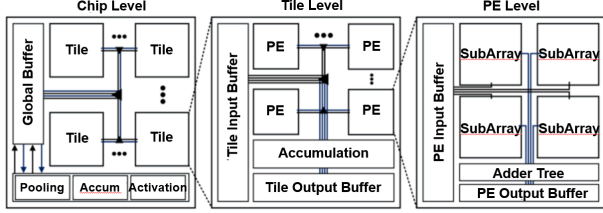
FIGURE 3. Designed subarray configuration.

input decimal mapping that will be introduced in Section III) requires relatively long (hundreds of ns) stabilizing time because of the large capacitance (~ 16 pF) of the SL of the 3-D NAND array. Although WL setup requires 43.5 nJ per each time, it needs to be conducted only once for each of the DNN layer, so the ratio over the total energy of WL setup energy is only 1%–5%.

Fig. 3 shows the subarray configuration with 3-D NAND array and peripheral circuits. Throughout this work, we assumed that the 3-D NAND array cells and peripheral logic transistors are fabricated on the same wafer. Alternative hybrid bonding scheme where the 3-D NAND die and logic die are integrated could be referred to [14]. Different from the conventional 3-D NAND architectures using low-cost peripheral transistors (hundreds of nm gate length), we adopted advanced logic transistors with 3-D NAND to reduce the peripheral area, except for the high-voltage transistors for WL voltage. This is mandatory because our relatively small ($64 \text{ blocks} \times 13,824 \text{ BLs}$) 3-D NAND subarray design needs a larger portion of total chip area to be dedicated for the peripheral circuits than conventional 3-D NAND array of which the array size is $>1000 \text{ blocks} \times 16 \text{ kB BLs}$. Input vectors are fetched to the BLs through the BL switch matrix, and XDEC enables the operating blocks. WL voltages are applied to the 3-D NAND of the selected blocks through the pass transistors. The summed current flowing through the SL is sensed at the edge and converted to digital value through ADC. Here, 7-bit ADC is used to guarantee the computational accuracy considering the large number of BLs summation along with the SL, as reported in [8]. In our design, the SAR-ADC is selected over the flash-ADC due to the relatively high resolution required here to trade off the latency over the power and area [15]. VMM of multicycle input vectors and the 2-bit weight produces multiple 7-bit ADC outputs. They are summed in the digital adder and shifters to achieve the partial sum results. The partial sums from the multiple blocks storing other significant weight bits are accumulated to achieve the

TABLE 2. Area estimation of the subarray modules with 32-nm technology for peripheral circuits.

Module	Area [mm ²]	Module	Area [mm ²]
BL switch matrix	0.028	ADC	0.0158
XDEC and driver	0.0014	Adder/shifter	0.0084
Pass transistors	0.0035	NAND array	0.0797
WL staircase	0.0014		

**FIGURE 4.** Hierarchy of the 3-D NAND-based CIM chip architecture.

final output. The MSBs of the final output that matches the weight precision are chosen to be fed into the next layer as the input. The estimated areas of each module for 32-nm peripheral logic technology nodes are shown in Table 2. The areas of the pass transistors and XDEC drivers are fixed because we assumed the high-voltage transistor process (1- μ m pitch) is applied here to drive the programming voltage (~ 20 V) to the WLs.

D. SUBARRAY-LEVEL WEIGHT COPY

Due to the high memory density of 3-D NAND, one subarray already has 81-Mb cells. Our chip hierarchy defines the minimum number of subarrays per PE and the minimum number of PEs per tile as four, so the minimum tile capacity is 1296 Mb, which is large enough to store today's large DNNs, such as VGG-16 (138 MB with 8-bit weight). Our selected VGG-8 model for the CIFAR-10 dataset contains 12.37-MB weight, so we can copy the weights to the other subarrays, PEs, and tiles even after MSB weight decimal mapping and multibit decimal mapping. How many times to copy the subarray (or PE, tile) depends on the targeted total chip size. We can compute the different input vectors in parallel with such weight copy. The mapping schemes are also compatible with larger and deeper network models, such as ResNet families [16], as will be discussed in Section V.

E. CHIP-LEVEL ARCHITECTURE DESIGN

The top-down hierarchy of the proposed 3-D NAND-based CIM architecture is defined as chip, tile, processing element (PE), and subarray, as shown in Fig. 4. The chip-level consists of tiles, global buffer, and neural functional computation logics for max pooling, activation (ReLU), and accumulation. The tile level is composed of several PEs, input/output buffers, and accumulation modules. Similarly, the PE levels are built with multiple subarrays, PE input/output buffers, and accumulation modules. The H-tree-based routing is used for the output accumulation at each level. The number of subarrays per PE and the number of PEs per tile can vary by optimization with respect to the topologies of various DNNs.

TABLE 3. Area estimation of the subarray modules with 32-nm technology for peripheral circuits.

Module	Value	Module	Value
Number of Tiles	4	Interconnect [mm ²]	2.11
Total Area [mm ²]	17.91	ADC [mm ²]	1.26
Cell Array [mm ²]	5.10	Accum [mm ²]	2.33
Cell Array Efficiency	28.5%	Other [mm ²]	7.08
Chip Capacity	1.13 Gb		

III. BENCHMARKING RESULTS

We defined the 3-D NAND subarray dimension and ON/OFF current, as discussed in Section II; then, the widely used open-source simulator DNN+NeuroSim [11] version 1.2 was customized as a benchmarking framework to estimate the area, memory utilization, latency, and energy consumption of the designed 3-D NAND-based CIM architecture.

Table 3 shows the designed chip area and breakdown result with 32-nm logic transistor technology nodes. Because we limited the total chip size under 20 mm² in this estimation, the number of tiles and total chip capacity varies with the technology node. The 32WL 3-D NAND process with a 32-nm logic process has 28.5% cell array efficiency, while the 7-nm logic process has 78.4%.

Basically, we mapped the weight of the one layer of DNN to the one WL of 3-D NAND unless the number of layers is larger than the number of total WLs. Table 4 shows the layer-by-layer weight mapping strategy of the VGG-8 network on the designed architecture with a 32-nm technology node. The shallow layers can employ higher bit input decimal mapping because the kernel size of the layers is small. Layer 1 has 128 kernels, which means that 512 blocks (eight subarrays) are required to store 8-bit weight. Consequently, layer 1 can be copied to the eight different subarrays in the 32-nm technology node. Each subarray computes different input feature maps (IFMs) in parallel, so eight times faster computing is possible in total.

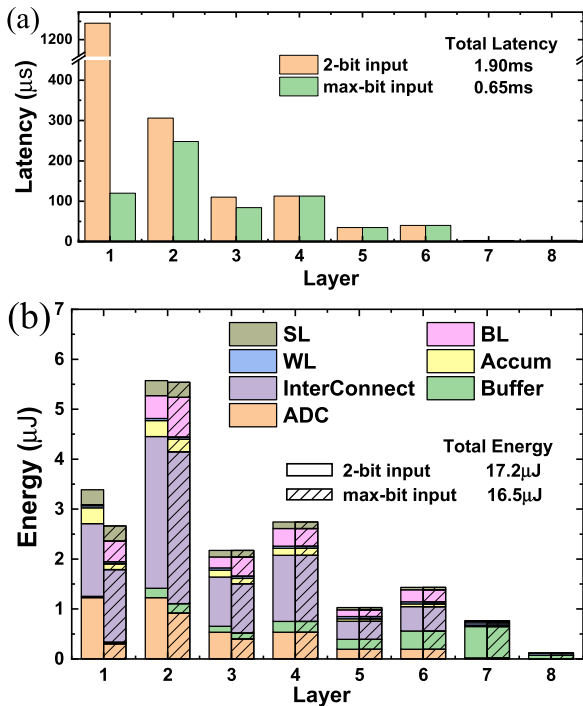
In terms of the max-bit input decimal mapping, layer 1 has only 27 of kernel size, so the weights can be mapped to $27 \times (2^8 - 1)$ BLs for 8-bit input decimal mapping. The 8-bit input can be computed in only one cycle. Layers 2 and 3 are mapped to seven times more BLs (3-bit input and three cycles), and other layers are mapped to three times more BLs (2-bit input and four cycles). The shallow layers have a large IFM size so that they can fully take the advantage of decimal mapping in terms of latency improvement. Finally, total decimal mapped weights are stored in 110.25-MB cells, which is 8.9 times larger than the original size (12.37 MB).

The memory utilization of each layer is also shown in Table 4. Since the subarray-level weight copy method employs whole blocks of the chip, 100% of the blocks are utilized irrespective of the kernel size of the layer. On the other hand, the number of activated BLs always depends on the kernel size and multibit input decimal mapping, so it is the key factor for memory utilization. For instance, layer 4 activates 6912 BLs in the subarray of the entire chip; then, the memory utilization is 50% as same as the number of activated BLs (6912) over the number of BLs in subarray (13 824).

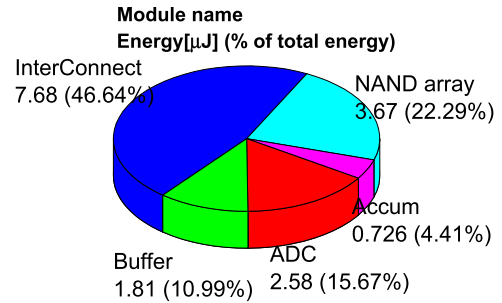
Fig. 5 shows the simulated layer-by-layer latency [see Fig. 5(a)] and energy consumption [see Fig. 5(b)] on the

TABLE 4. Weight mapping strategy and speed up of VGG-8 on total chip (32 nm) and memory utilization of each layer.

	Weight size (8-bit weight precision)	Subarray weight copy	max-bit input decimal mapping	MSB weight decimal mapping	1cell is copied	Required number of cells	Memory Utilization of each layer	Total Speed-up
Layer1	27 B × 128	8×	255× (8bit)	1.5×	3060×	10.09 MB	49.8%	128×
Layer2	1152 B × 128	8×	7× (3bit)	1.5×	84×	11.81 MB	58.3%	48×
Layer3	1152 B × 256	4×	7× (3bit)	1.5×	42×	11.81 MB	58.3%	24×
Layer4	2304 B × 256	4×	3× (2bit)	1.5×	18×	10.12 MB	50.0%	16×
Layer5	2304 B × 512	2×	3× (2bit)	1.5×	9×	10.12 MB	50.0%	8×
Layer6	4608 B × 512	2×	3× (2bit)	1.5×	9×	20.25 MB	100.0%	8×
Layer7	8192 B × 1024	1×	3× (2bit)	1.5×	4.5×	36 MB	88.9% (2WL)	4×
Layer8	1024 B × 10	1×	3× (2bit)	1.5×	4.5×	0.044 MB	22.2%	4×
Total	12.37MB					110.25 MB	62.94%	


FIGURE 5. Simulated layer-by-layer (a) latency and (b) energy result on the VGG-8 network for the CIFAR-10 dataset.

VGG-8 network for the CIFAR-10 dataset with a uniform 2-bit input decimal mapping and an adaptive max-bit input decimal mapping. The shallow layers cause slow latency and large energy consumption because their large IFM size needs many times of computations. The max-bit input decimal mapping can not only significantly reduce the latency of the first layer (8-bit input) but also decrease the energy consumption. The number of activated BLs is 85 times larger than that of 2-bit input decimal mapping, so the energy consumed in the single-input cycle for charging the BLs and BL switch matrix increases as the mapped number of BLs increases, from 0.64 to 54.5 pJ, respectively. However, the energy consumption within other peripheral modules (e.g., multilevel


FIGURE 6. Energy consumption breakdown of 3-D NAND-based accelerator on VGG-8 network for the CIFAR-10 dataset with max-bit decimal mapping. Accum logic contains accumulation, max pooling, and activation logic.
TABLE 5. Benchmark results of various DNN accelerators on the VGG-8 model for the CIFAR-10 dataset.

Technology node	32nm	32nm	32nm
Device	SRAM	RRAM	3D NAND
ADC Precision	4-bit	5-bit	7-bit
Cell Precision	1-bit	2-bit	1-bit
R _{on} (ohm)	-	6k	100M
On/Off Ratio	-	17	2000
Area (mm ²)	103.49	80.12	17.91
Memory Utilization (%)	94.38%	91.13%	33.50%
Latency (ms)	17.04	27.98	5.07
Dynamic Energy (uJ)	297.8	290.0	138
Leakage power (mW)	2.34	0.60	0.12
Energy Efficiency (TOPS/W)	6.89	7.27	12.95
Throughput (FPS)	58.68	35.74	197.24

* This memory utilization is calculated at used WLs only. ** Percentage in whole WLs (32WLs).

sense amplifier: 42.6 pJ) does not change in the single-input cycle. Therefore, the subarray energy consumption in the single-input cycle increases, but the number of input cycles decreases four times less. In conclusion, the chip-level energy consumption of the first layer decreases by 21.6%, and total energy consumption decreases by 5%.

Fig. 6 shows the breakdown of the energy consumption of the max-bit input decimal mapping case. Due to the multibit input decimal mapping, which can effectively reduce the

TABLE 6. Benchmarking result of 3-D NAND-CIM architecture on ResNet-18 to ResNet-152 for the ImageNet dataset. The chip area is fixed to 17.91 mm².

	ResNet-18	ResNet-34	ResNet-50	ResNet-101	ResNet-152
ADC Precision			7-bit		
Cell Precision			1-bit		
Weight Parameter Size (8-bit)	11.5 MB	21.6 MB	22.7 MB	41.7 MB	48.6 MB
Memory Utilization	33.5%	63.5%	68.9%	98.6%	99.2%
Latency (ms)	5.07	7.93	21.7	40.3	61.7
Energy (uJ)	138	203	316	510	740
Leakage (mW)	0.12	0.12	0.12	0.12	0.12
Operation (GOPs)	1.80	3.64	3.50	7.21	10.9
TOPS/W	12.95	17.87	10.99	14.00	14.61
Throughput	197.24	126.10	46.08	24.81	16.21

number of NAND array operations, NAND array only consumes 22.29% of total energy, while 46.64% is consumed in interconnect buses in H-tree routing. It implies that input/output energy is also critical for CIM chip design, so further optimization of the interconnect bus and the input/output feature map dataflow is essential for future work to achieve even higher energy efficiency.

Table 5 shows the benchmarking result comparison across the state-of-the-art device technologies. DNN+NeuroSim was also used to evaluate the other technologies: SRAM and RRAM. Evaluated SRAM- and RRAM-based CIMs have 128×128 cell array size [11]. Although all the decimal mapping and weight copy methods are applied on 3-D NAND array, the total chip size is only 17.91 mm², which is significantly smaller than the CIM chips with other technologies (73.53 mm² for RRAM and over 100 mm² for SRAM). Due to the multiple mapping methods in the 3-D NAND array, the 3-D NAND flash-based accelerator has shown the 1545.6 frame per second (FPS), which is the highest throughput compared to the SRAM and RRAM technology at the same 32-nm technology node with the low-standby power (LSTP) library.

For SRAM, there are two designs: one is the conventional row-by-row readout with the digital adders at the edge of the array to perform near-memory compute, and the other is the SRAM-CIM mode where the peripheral circuits are modified to support parallel read-out in SRAM array, as recently demonstrated in several silicon prototypes [17], [18]. Originated from its large R_{on} in the 3-D NAND array, the large number of BL inputs results in the huge parallelism with 37.1 TOPS/W energy efficiency, outperforming other technologies at the same technology node. It is noted that the TOPS/W reported here is for 8-bit input by 8-bit weight MAC operations, and it is equivalent to 148.4 TOPS/W for 4-bit input by 4-bit weight operations if using the same precision definitions in some of the prior works. For example, the 28-nm SRAM-CIM macro reported 68 TOPS/W [18]; and the 22-nm RRAM-CIM macro reported 29 TOPS/W [19]. Our proposed 3-D NAND-CIM design showed 2.2–5.1 times improvement in energy efficiency to state-of-the-art CIM designs.

IV. DEEPER NETWORK BENCHMARKING FOR 3-D NAND

As explained in the previous sections, we mapped each layer of the VGG-8 network to the single WL (except layer 7 that is the largest FC layer that does not fit into a single WL).

Because multiple WLs cannot be read at the same time in the 3-D NAND array, coping the weights to other WLs (WL10-32) does not have any advantage for the speed up, so the large number of WLs remain empty and have not been fully exploited on the relatively shallow DNNs.

Therefore, we have considered the deeper network, such as the ResNet [16] families, for the ImageNet dataset to be mapped to the 3-D NAND-based CIM. First, we mapped our 32WL 3-D NAND CIM on the various ResNet networks, from ResNet-18 to ResNet-152. The deeper the network is, the more WLs can be exploited, so we could achieve higher memory utilization of the whole 3-D NAND array, as shown in Table 6. Because ResNet-34 to ResNet-152 have more than 32 layers, some (or whole) of the WLs store the weight of more than two layers. As can be seen in the memory utilization in Table 6, ResNet-18 to ResNet-50 have enough storage space to fully utilize max-bit input decimal mapping. However, ResNet-101 and 152 cannot utilize the max-bit input decimal mapping for the whole layer, so we optimized the number of input decimal mapping layer by layer. The layers with large kernel size ($3 \times 3 \times 512$ and $1 \times 1 \times 1024$) were not mapped to other BLs for multibit input decimal mapping.

Irrespective of the number of layers in the network, the energy efficiency on ResNet is relatively lower than that on VGG-8. The energy efficiency of the individual layer is strongly related to the kernel size. Regardless of the kernel size, the whole 3-D NAND array and the 7-bit ADC should operate once for one weighted sum computation without significant variation in energy consumption. Therefore, the small convolution kernels, such as $3 \times 3 \times 64$ in the shallow layers and 1×1 convolutional kernels (only in ResNet-50, ResNet-101, and ResNet-152), tend to have low energy efficiency. Not only the 3-D NAND-based design, other technology-based designs also have lower energy efficiency on ResNet (see Table 7) over VGG-8 for the same reason.

Second, we evaluated the area and performance on ResNet-18 for the ImageNet dataset with various device technologies the same as the VGG-8 case, as shown in Table 7. Because 3-D NAND-based CIM could utilize more WLs and copy the weights more than the case of VGG-8, the throughput is 3.4 and 5.5 times higher than that of SRAM and RRAM, respectively (1.2 and 1.9 times higher for VGG-8). We could also achieve the energy efficiency, which is 87% and 78% higher than that of SRAM and RRAM. Again, the 3-D NAND chip area is just 17.91 mm², but the SRAM-CIM chips are exploded to over 100 mm².

TABLE 7. Benchmarking result of SRAM, RRAM, 3-D NAND-based DNN accelerators on ResNet-18 for the ImageNet dataset. The 8-bit input and 8-bit weight precision were used.

Technology node	32nm	32nm	32nm
Device	SRAM	RRAM	3D NAND
ADC Precision	4-bit	5-bit	7-bit
Cell Precision	1-bit	2-bit	1-bit
R _{on} (ohm)	-	6k	100M
On/Off Ratio	-	17	2000
Area (mm ²)	103.49	80.12	17.91
Memory Utilization (%)	94.38%	91.13%	33.50%
Latency (ms)	17.04	27.98	5.07
Dynamic Energy (uJ)	297.8	290.0	138
Leakage power (mW)	2.34	0.60	0.12
Energy Efficiency (TOPS/W)	6.89	7.27	12.95
Throughput (FPS)	58.68	35.74	197.24

* This memory utilization is calculated at used WLs only. ** Percentage in whole WLs (32WLs).

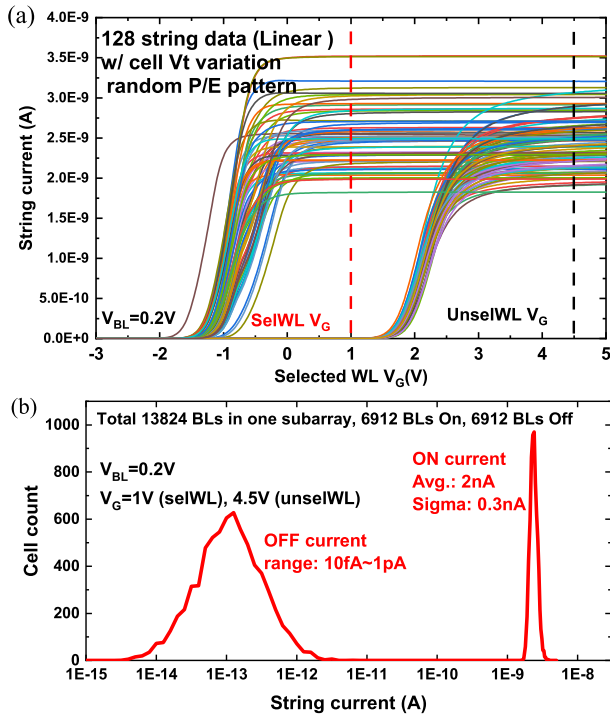


FIGURE 7. NAND string current variation with cell V_{th} variation and back pattern dependency. (a) I_D - V_G of 128 strings with random program/erase pattern. (b) On and off cells' distributions.

V. IMPACT OF NONIDEAL CELL CHARACTERISTICS ON INFERENCE ACCURACY

Although the CIM designs show improved energy efficiency compared to the digital computing architectures, the inference accuracy of the DNN models needs to be investigated because it may be vulnerable to the variation of the memory cells' read current. Any variation of the read current may be aggregated since the current is summed up along the column in an analog manner, resulting in the ADC quantization error. We investigated the effect of the current variation of 32 WL NAND strings on the inference accuracy in our design.

For the first step, to evaluate the NAND string current variation using HSPICE simulation, the BSIM model is modified to fit the flash transistor's I - V experimental data in [8].

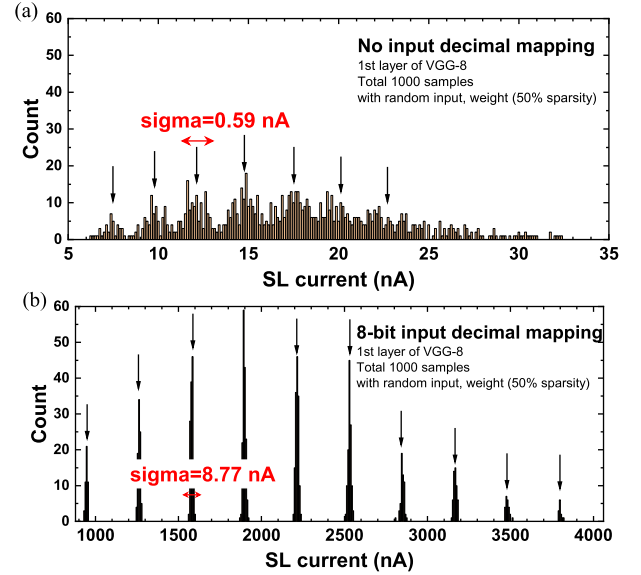


FIGURE 8. NAND string current distribution (a) without input decimal mapping and (b) with max-bit (8-bit) input decimal mapping.

We added two major sources of the variation on the 3-D NAND netlist, the threshold voltage variation of an individual cell, and the back pattern dependency (i.e., current variation caused by the program/erase state of the unselected WLs). We set the standard deviation of V_{th} of on cells (low V_{th}) and off cells (high V_{th}) to 0.12 and 0.05 V, respectively. Fig. 7(a) shows the I_D - V_G curve of the 128 NAND string with two current variation sources. Not only V_{th} of each individual string has the variation but also the saturated ON-current ($V_G > 1$ V) has the variation, which may cause the error in the quantization of summed current. We simulated 13 824 BL NAND array at read bias condition and extracted ON- and OFF-current distributions, as shown in Fig. 7(b).

Next, we evaluated the advantage of max-bit input decimal mapping in the viewpoint of the summed current variation. We simulated the max-bit input decimal mapping scheme by coping the 50% sparse random input vectors and weights to the BLs. Then, the total summed current (SL current) distributions of 1000 samples are observed. As shown in Fig. 8, the max-bit input decimal mapping can reduce the quantization error on the summed current. Although the absolute value of the SL current variation (σ) in the first layer of VGG-8 increases from 0.59 to 8.77 nA, the variation ratio over the total summed current (σ/μ) was reduced ten times from ~ 0.04 to ~ 0.004 due to the central limit theorem.

Finally, the current variations of all the layers in VGG-8 extracted from the HSPICE simulation were incorporated into the software simulation. After we considered the ADC quantization effect on the ResNet-18 model, the inference baseline top-1 accuracy was only 67% on the ImageNet dataset, so we have only considered the device nonideal effects on VGG-8 in this work. The VGG-8 model was pretrained with the CIFAR-10 dataset by software with the WAGE quantization method [20]; 90.3% of accuracy is achieved after the 257 epoch of the training phase. At the inference phase, 8-bit inputs and weights are quantized and then mapped to eight

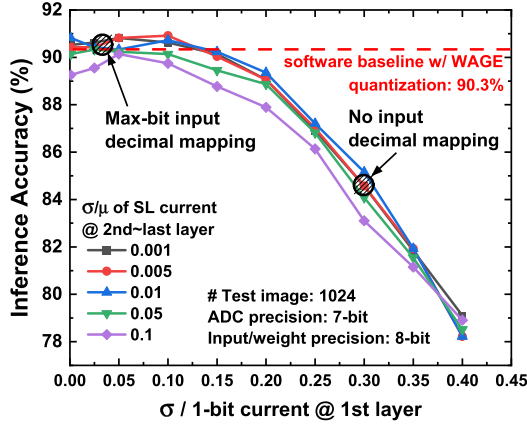


FIGURE 9. Simulated CIFAR-10 inference accuracy with various SL current variations.

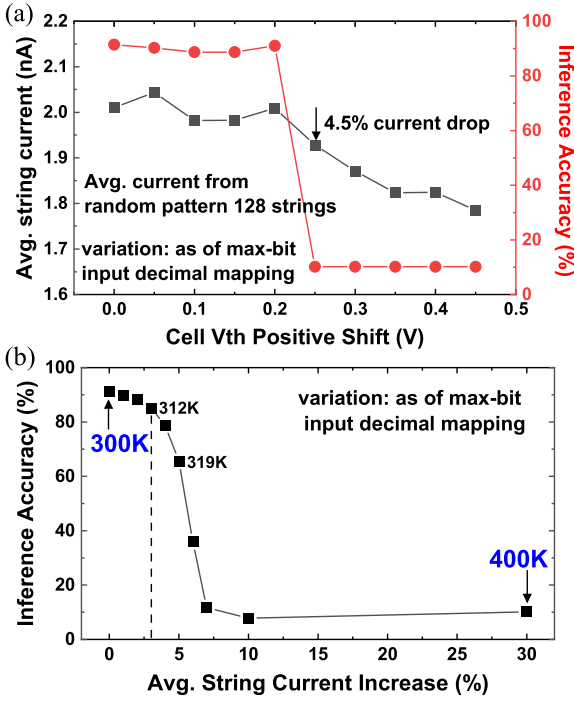


FIGURE 10. (a) Inference accuracy drop induced by average string current decrease due to positive threshold voltage shift (current drop) at retention case. (b) Inference accuracy drop induced by string current increase with temperature increase. The X-axis is the increase ratio over the initial string current.

cells. The standard deviation of SL currents was added to the convolution results of inputs and weights. Then, the SL current is quantized to 7-bit and digitized. The inference accuracy was measured by testing 1024 images.

Fig. 9 shows the inference accuracy with various SL current variations. Without cell current variation, the inference accuracy is maintained at software baseline with 7-bit ADC precision. Because the SL current variation on the first layer of VGG-8 drastically drops the inference accuracy, we tested with separated standard deviation parameters for the first layer and other layers. The x-axis represents the standard deviation divided by the 1-bit current (quantized distance in Fig. 8) of the first layer. The other parameter is the standard

deviation over the average SL current of the other layers where each of the other layers has a similar σ/μ value. Without input decimal mapping, a relatively large variation of the first layer (0.59 over 2 nA) induces an accuracy drop to 84.5%, while the max-bit decimal mapping variation (8.77 nA over ~ 2500 nA) does not induce the accuracy drop. The other layers have ~ 0.005 (no input decimal mapping) or ~ 0.003 (max-bit input decimal mapping) of σ/μ , which does not induce the accuracy degradation.

Other than the summed current variation, 3-D NAND memory cells have two major sources drifting the average summed current. First, the retention time may cause threshold voltage shift, which induces the current drop of ON-state cells because of the charge (electron) gain into the charge trapping layer of low threshold voltage cells. The OFF-state (high threshold voltage) cells may have a current increase due to the charge loss, but the effect on the total current is ignorable due to the high on and off ratio. Therefore, we have only considered the ON-current drop in the following results. Referred from [21], 3-D NAND cells generally have <0.4 V threshold shift up to 1000-h retention time at 85 °C, and the lower threshold voltage states tend to be affected less than the higher threshold voltage states. To investigate the current decrease with the HSPICE simulation, we incorporated the threshold voltage shift into our BSIM model. Because all the ON-state cells within a string are affected, the average string current drops up to 9.5% at 0.40-V shift, as shown in Fig. 10(a). The string current drop was incorporated into the DNN+Neurosim simulator to evaluate the effect on the inference accuracy. As a result, the inference accuracy was maintained near the baseline accuracy up to 0.2-V threshold voltage shift. However, the accuracy drops abruptly with a 0.25-V shift where the current drops 4.5% than the initial state. It can be noted that ~ 0.2 -V threshold voltage shift would be the reprogram criteria of 3-D NAND cells to maintain the inference accuracy level.

Second, the temperature variation will also cause the change of the average summed current. The mobility of the polysilicon channel of 3-D NAND increases with the temperature at a read operation. According to the [22], the mobility of the polysilicon channel increases by 30% when the temperature varies from 300 K to 400 K. Hence, we investigated the effect of the current increase of up to 30% on the inference accuracy, as shown in Fig. 10(b). Since we have not applied any compensation schemes to the temperature effect, the inference accuracy drops quite sensitive to the current increase. These results imply that, in the practical circuit design, the reference current adjustment (e.g., by dummy cells) to compensate for the temperature effect must be implemented for the accurate CIM applications.

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WONBO SHIM received the B.S. and Ph.D. degrees in electrical engineering from Seoul National University, Seoul, South Korea, in 2007 and 2013, respectively.

From 2013 to 2019, he was a Senior Engineer with the Flash Design Team, Memory Division, Samsung Electronics, Hwaseong, South Korea. Since 2019, he has been a Post-Doctoral Research Fellow with the Georgia Institute of Technology, Atlanta, GA, USA. His current research interests include energy-efficient deep learning architecture design, nonvolatile memory device modeling for synaptic applications, and 3-D NAND flash-based compute-in-memory design.

SHIMENG YU (Senior Member, IEEE) received the B.S. degree in microelectronics from Peking University, Beijing, China, in 2009, and the M.S. and Ph.D. degrees in electrical engineering from Stanford University, Stanford, CA, USA, in 2011 and 2013, respectively.

From 2013 to 2018, he was an Assistant Professor of electrical and computer engineering with Arizona State University, Tempe, AZ, USA. He is currently an Associate Professor of electrical and computer engineering with the Georgia Institute of Technology, Atlanta, GA, USA. His research interests are nanoelectronic devices and circuits for energy-efficient computing systems. His expertise is on the emerging nonvolatile memories (e.g., resistive random access memory (RRAM) and ferroelectrics) for different applications, such as machine/deep learning accelerator, neuromorphic computing, monolithic 3-D integration, and hardware security.

Dr. Yu was a recipient of the NSF Faculty Early CAREER Award in 2016, the IEEE Electron Devices Society (EDS) Early Career Award in 2017, the ACM Special Interests Group on Design Automation (SIGDA) Outstanding New Faculty Award in 2018, the Semiconductor Research Corporation (SRC) Young Faculty Award in 2019, the ACM/IEEE Design Automation Conference (DAC) Under-40 Innovators Award in 2020, and so on.

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